

Author Response Letter

Dear Reviewers,

We thank the reviewers for their thoughtful and constructive feedback on our ICRA 2026 submission. We have carefully revised the manuscript to address all concerns, with particular focus on clarifying novelty, strengthening experimental validation through an ablation study, and demonstrating real-time feasibility on a wearable exoskeleton.

Major Revisions and Improvements -

1. Clarifying Novelty:

- Restructured the paper to focus on wearable robotics applications.
- Emphasized the core novelty of our work: **a phase conditioned latent space for long-horizon joint kinematic and kinetic forecasting from realistic wearable IMU data**, integrated with real-time deployment on a wearable exoskeleton—a combination that has not been demonstrated in prior work.

2. Evaluation and Baseline Comparison:

- Expanded our evaluation to include multiple baselines, including fully-connected, temporal convolution networks and recurrent neural networks.
- Included **task-specific metrics such as RMSE and correlation of predicted angles and moments**, which are more appropriate than MPJPE for joint-level kinematic and kinetic prediction.

3. Emphasizing the importance of phase:

- Added an ablation study showing that removing phase conditioning or expert-weight gating significantly degrades long-horizon performance.
- Highlights the necessity of the phase-conditioned latent space for capturing temporal progression in locomotion.

4. Addressing Dataset Size / Generalization Concerns

- Performed extensive leave one subject out evaluation to demonstrate model robustness across multiple locomotion tasks.
- Developed a **custom wearable sensor suit** for future collection of larger and more diverse datasets, which our methodology supports.

5. Real-Time Deployment/Feasibility:

- Deployed the model on a **wearable hip-knee exoskeleton**, achieving an **average inference time of 3.7ms with total system latency under 11 ms**, demonstrating the suitability for online control.
- Included a video demonstration showing the system operating in real time.

6. Manuscript Clarity and Presentation:

- Revised the text for improved clarity, flow, and presentation.
- Reorganized figures and tables, highlighting key contributions and results to make the manuscript more accessible.

We believe these revisions thoroughly address the reviewers' concerns regarding novelty, experimental rigor, and real-time feasibility, and we hope the manuscript is now suitable for acceptance.

Sincerely,
The Authors